

Initial Analysis of Monroe County Crashes, 2012-2024

Group 1: DSCC 383W, Goergen Institute for Data Science
University of Rochester

1. Introduction

Automobile crashes continue to be a major issue plaguing America's roadways, with a total of 39,254 killed and 2,422,195 injured in 2024 [1]. In addition, the COVID-19 pandemic saw increases in road fatalities even as overall crashes declined [2], especially with regards to pedestrians [3]. As a whole, American road safety remains poor compared to other developed countries, with the United States seeing 11.6 deaths from road crashes per 100,000 individuals, compared to 1.6 in Norway and 2.4 in the U.K and Japan [4].

Within this context, our sponsor, Reconnect Rochester, advocates for comprehensive solutions to the problem of transit, especially in the local context of Monroe County [5]. To support their case, Reconnect Rochester has come into possession of crash data from 2012-2024 for the Monroe County region, and wanted our assistance in extracting statistical findings in four key areas: the role of the **time of day** in crashes, the role of **differing jurisdictions** in crashes, **specific areas (corridors)** in which crashes occur, and the effectiveness of various **road projects** in preventing crashes. Due to national-level trends [3], a special emphasis was placed by Reconnect Rochester on **vulnerable road users** – defined as individuals who do not use cars for transportation (eg: pedestrians, cyclists, skaters, skateboarders) – and on looking at **fatal and serious injuries** (FSIs) rather than total crash counts.

2. Data & Data Cleaning

Our initial merged data consisted of approximately 291634 crashes from the years 2012 to 2024, across 3908 roads. These entries consisted of 65 variables, a detailed description of which can be found in the attached data dictionary (Data_Dictionary_for_Crash_Table.pdf).

The data was gathered by the NYS Department of Transportation in their 'CLEAR' database, which collects reports made by local police departments, and then later released to Reconnect Rochester via a Freedom of Information Act Request. This means that the data, by its nature, excludes any crashes which were not reported, or tend to be under-reported to the police, including, for example, crashes involving pedestrians and bicyclists.

The dataset has a number of other issues: the most prominent of these issues is the fact that street names in the variables 'OnStreet' and 'ClosestCrossStreet' are inconsistently named.

Data Merging

The original dataset was given to us in the form of 6 separate files: 2012-13_Crash_Events.csv, 2014-20_Crash_Events.csv, 2021_Crash_Events.xlsx, 2022_Crash_Events.xlsx, 2023_Crash_Events.csv, 2024_Crash_Events.xlsx. This was merged into a single dataset – 2012-24_Crash_Events.csv.

Street Name Standardization

In order to correct various misspellings and inconsistencies with regards to road names, the variables 'OnStreet' and 'ClosestCrossStreet' (which represent the street a crash occurred on, as well as the closest street nearby) were renamed using a combination of string substitution and regex. We created a function (with assistance from LLMs to save on time) to substitute abbreviated parts of roads (eg: 'AVE.' was

replaced with AVENUE), remove directional terms (“Northbound / Southbound”), and remove mentions of ramps (“Ramp to Interstate 490”). We then created a second function to handle inputted street names that were simply numbers (eg: “490” was made into “Interstate 490”), and created code to handle the special case of Interstate / State Route 390, based on the variable ‘UTMNorthing’.

The cleaned names were then saved to new variables named ‘OnStreet_standardized’ and ‘ClosestCrossStreet_standardized’; the modified dataset was then saved as ‘2012-24_Crash_Events_std_street_names_dataset.csv’.

Table 2: Examples of street name cleaning

BEFORE	AFTER
Lake Av. Northbound	LAKE AVENUE
259	STATE ROUTE 259
390	INTERSTATE 390 (if UTM-North coordinates were less than 4782111.55) STATE ROUTE 390 (if UTM-North coordinates were greater than or equal to 4782111.55)

Fatal and Serious Injuries:

Reconnect Rochester specifically requested that we focus on fatal and serious injuries as our primary metric. This is for two key reasons; first, this is meant to align with Reconnect Rochester’s support for the City of Rochester’s Vision Zero policy [6]. Second, although traffic crashes decreased during COVID, traffic deaths, especially among pedestrians and bikers, spiked [2]. As a result, for most of our analysis, fatal injuries, serious injuries, or a combination of the two are the basic metric of analysis.

Determining the number of fatal and serious injuries consists of looking at the variable “NumberOfSeriousInjuries” and “NumberOfFatalInjuries”. From this, we can see that 5462 (1.9%) resulted in at least one serious injury & 559 (0.2%) resulted in at least one fatality, for a total of 6219 people seriously injured & 585 killed by cars from 2012-2024.

Vulnerable Road Users

In order to focus on vulnerable road users, we created a vulnerable-road-user specific dataset named pedestrian_cyclist_collisions_all.csv, which relied on looking at the variable “CrashType” and extracting entries which had pedestrians or bikers listed, using regex (r"PEDESTRIAN|BICYCL"). This yielded 6997 entries, of which 1133 were fatalities or serious injuries.

ArcGIS Data

We also have ArcGIS data from the NYS DOT Roadway Inventory [7], an ArcGIS database of all roads within New York State, and extracted all roads within Monroe County using the variable “County_Name”. We then ran the same code we used for Street Name Standardization on “Roadway_Name”, before merging the ArcGIS data with ‘2012-24_Crash_Events_std_street_names_dataset.csv’. We discarded roads not in “2012-24_Crash_Events_std_street_names_dataset.csv”, and backfilled the gaps by manually selecting

missing segments from the original ArcGIS file and merging them with the new shapefile – a collection of vector layers on the map used for roads.

3. Time of Day Analysis

Time of day is a well-known factor in crashes & crash severity; however, an analysis has not been done, as far as we know, on time of day as a factor specifically for Monroe County crashes, in particular with regards to fatal and serious injuries.

Exploratory data analysis (EDA)

As Figures 1 and 2, show, both the General Population and the Vulnerable Road User population show peaks at around 17:00 hours, as well as a smaller peak at 7:00 to 8:00. Increased crash rates at these ‘rush hours’, when people commute to and from work, are a well-documented phenomenon [8]. However, a look at FSI rates in figures 3 and 4 suggest that, when crashes do occur, they are far more likely to become fatal at night, when there is lower traffic.

Figure 1: General Population Total Crashes by Hour

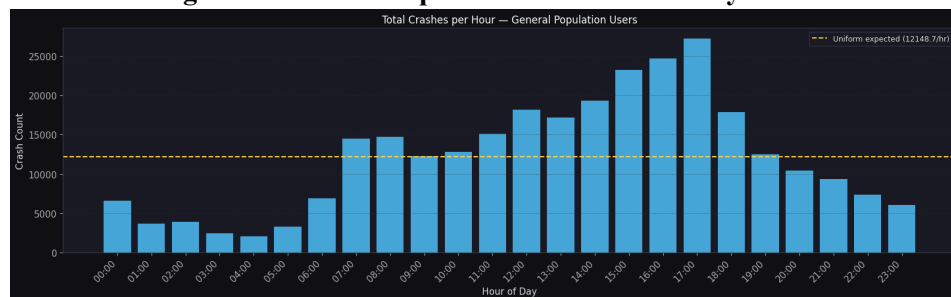


Figure 2: Vulnerable Road User Population Total Crashes by Hour

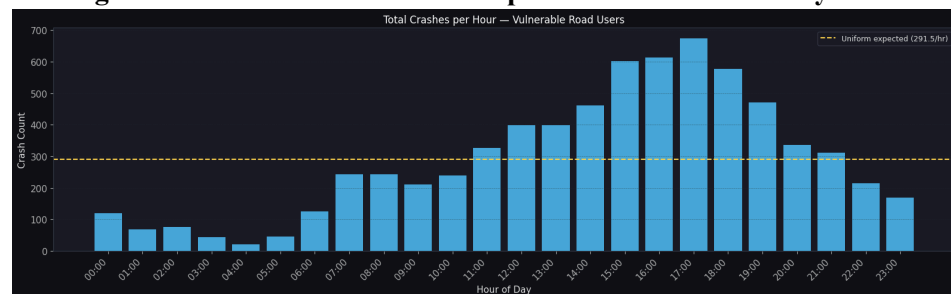


Figure 3: General Population FSI by Hour

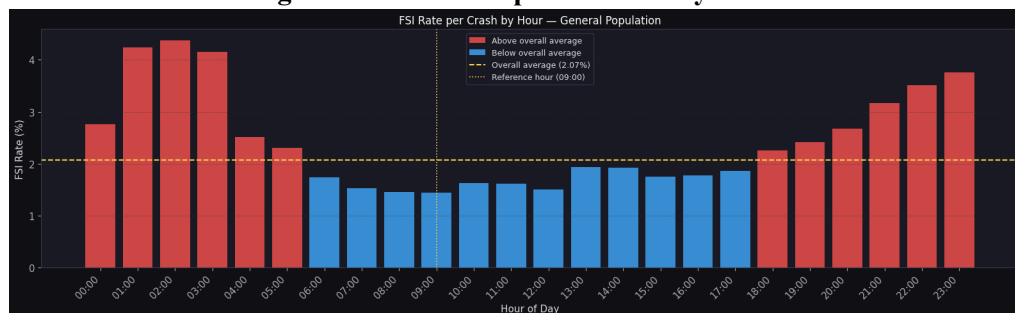
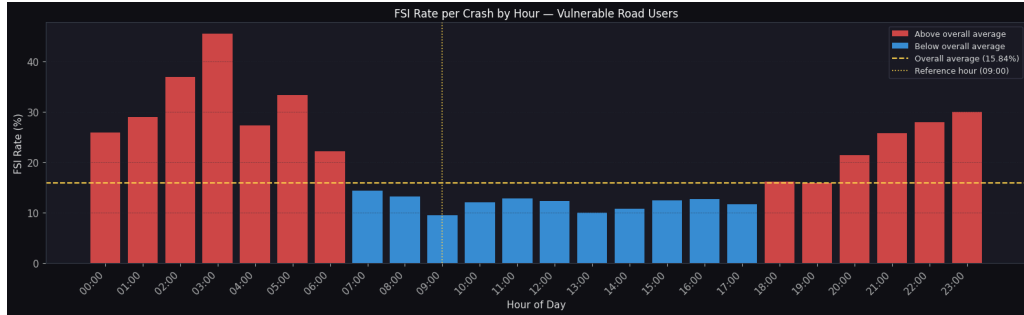


Figure 4: Vulnerable Road User FSI by Hour



Methodology

In order to confirm the heightened FSI rates at night / in the early morning, we applied a binary logistic regression to model the conditional probability of a crash resulting in a fatal or serious injury. The analysis was conducted on the full dataset as well as the vulnerable users dataset. The binary outcome was coded 1 if a crash resulted in fatality or serious injury and 0 if not. The hour of day was treated as a categorical predictor with 24 levels for the 24 hours and the reference category was set as 9:00am, the hour with the lowest FSI rate. The log-odds ratio of the hours was plotted for each hour to statistically determine how dangerous a given hour was in terms of fatality/serious injury.

Results

We can observe from Figures 5 and 6 that, by Log-Odds, the hours where both the General Population and Vulnerable Road Users both have higher FSI rates are *not* the same as the hours where the most crashes occur. Fatal and serious injuries have statistically significant odds of occurring *not* at peak hours, but late at night, between the hours of 18:00 to 0:500 for the General Population, and from 20:00 to 06:00 for vulnerable users (excluding 04:00).

Figure 5: General Population FSI Rate & FSI Log Odds by Hour

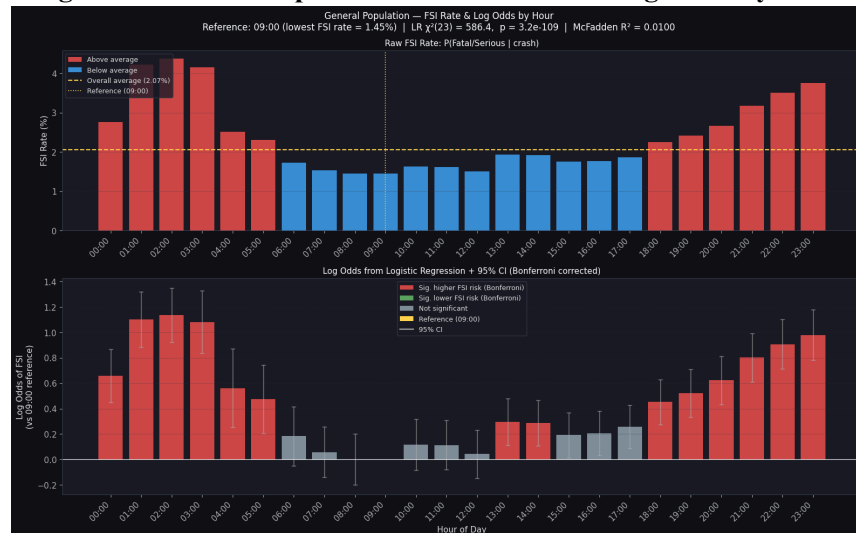
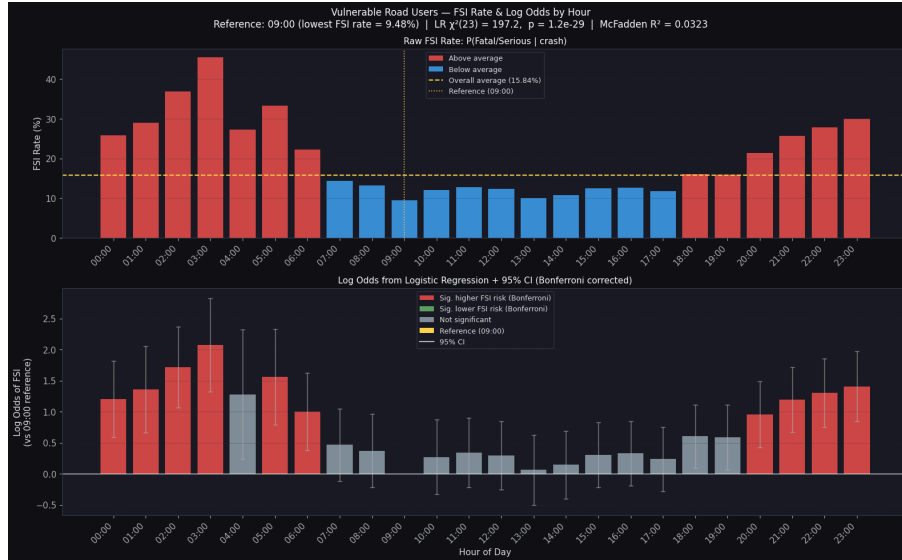


Figure 6: Vulnerable Road User Population FSI Rate & FSI Log Odds by Hour



4. Jurisdiction Analysis

Different roads in Monroe County are administered by different jurisdictions, with the primary jurisdictions being: various local towns, the City of Rochester, Monroe County, and the State Department of Transportation. Different jurisdictions build and maintain roads to different standards, with the NYS DOT usually building them to laxer standards. Reconnect Rochester therefore is interested in how different jurisdictions affect crash counts and fatal / serious injuries, especially for vulnerable road users.

EDA

We first looked at the FSI rate by Jurisdiction for the general population, shown in Figure 7. It shows that there is a nominal difference in FSI rates between jurisdictions.

Figure 7: General pop. FSI Rate by Jurisdiction

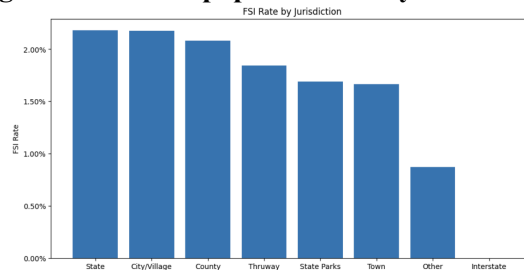


Figure 8: QQPlot, Serious Injuries

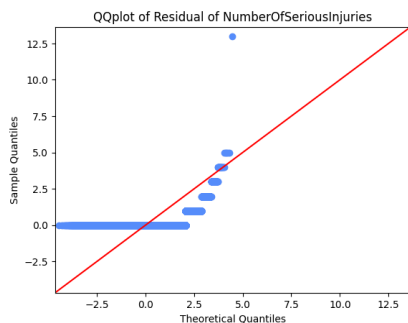
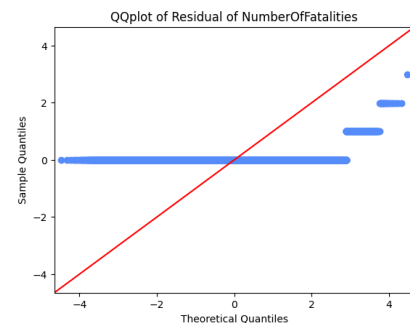


Figure 9: QQPlot, Fatalities



Figures 8 and 9 reveal that FSI counts are *not* normally distributed, which rules out parametric tests like ANOVA (supported by Shapiro test result of residuals of Serious Injuries and Fatalities with $p < 0.05$).

Methodology

As our data isn't normally distributed, we had to take a more complex approach which relied heavily on nonparametric methods – namely, the Kruskal-Wallis/Dunn Test [9] and the chi-squared test of independence (and the associated Cramer's V to see the effect size). Nonparametric tests, because they make no assumptions about the data distribution, are suited for non-normal distributions like crash and fatal/serious injury counts, though they also hold less statistical power [10].

For the general population, serious injury counts were transformed into an ordinal variable with three levels: 0, 1, and 2+, and certain features were removed (Interstate and Other due to relatively very small sample size). We then use the Kruskal-Wallis test to assess whether the distribution of injury severities differs across any jurisdictions, as well as a Dunn Test to determine which jurisdictions differed; we then used a chi-squared test to test of independence to test whether FSI crash outcome is associated with jurisdiction and then followed up with Cramér's V to determine the effect size, as well as pairwise proportion z-tests to better determine differences in FSI *rates* between jurisdictions.

For the vulnerable population analysis, we again used a chi-squared test of independence to see if the distribution of outcomes varied by jurisdiction; we then followed up with Cramér V to determine the effect size, and a pairwise proportion z-test with Fisher's exact test for jurisdiction with smaller sample size to determine which groups were different from each other. We also used a logistic regression model to estimate the log-odds ratio of FSI crashes for vulnerable road users (pedestrians) relative to the non-vulnerable populations (i.e, people driving cars).

Results – General Population

The Kruskal–Wallis test was used to assess whether the distribution of serious injury counts differed across jurisdictions with:

H0: The distribution of ordinal serious injury severity is identical across jurisdictions.

Ha: At least one jurisdiction differs in the distribution of ordinal serious injury severity.

The test resulted in an H-statistic of ~24.042 and a p-value of ~0.00021. These results indicate that the null hypothesis can be rejected, suggesting that at least one jurisdiction differs in the distribution of ordinal serious injury severity.

Therefore, a Dunn post hoc test with Holm adjustment was conducted to determine where the significant pairwise differences occurred with:

H0: The two jurisdictions have the same distribution of ordinal serious injury severity.

Ha: The two jurisdictions have different distributions of ordinal serious injury severity.

The Dunn test result graph (figure 10) showed that only a small number of jurisdiction pairs were statistically significant, indicating that although the overall difference across jurisdictions was significant, most pairwise differences were relatively small in magnitude. Statistically significant differences with regards to the count of serious injury crashes were between Towns and Cities / Villages and the State (alias NYSDOT) roads.

Likewise, a chi-squared test of independence between jurisdictions and fatal and serious injuries (alias FSI) outcome is done with:

H0: FSI outcome is independent of jurisdiction.
Ha: FSI outcome is not independent of jurisdiction.

The test returned a statistic of ~22.28, a p-value of ~0.00046, with 6 degrees of freedom and small effect size Cramér's $V = 0.009$. These results indicate a weak statistically significant association between jurisdiction and FSI outcome in practical terms.

Thus, A pairwise-proportion z-test on the FSI rate therefore were performed to compare FSI rates across jurisdictions, with Bonferroni correction applied to adjust for multiple comparisons:

H0: The two jurisdictions have the same FSI rate.
Ha: The two jurisdictions have different FSI rates.

The test result (figure 11) revealed that Towns Town roads differed significantly from City/Village, County, and State (alias NYSDOT) roads. And all other pairwise comparisons were not statistically significant after adjustment. Therefore, Town roads were the main jurisdiction type driving the overall difference in FSI rates. And as the difference heatmap of FSI rate (figure 12) shows the direction and magnitude of pairwise differences across jurisdiction. Positive values indicate higher FSI rate, and vice versa. As the result shows most of the differences are small, and Town had FSI rates only about 0.4 -- 0.5 percentage points higher than City/Village, County, and State, while differences involving State Parks and Thruway roads were also modest. These results suggest that, although some pairwise comparisons may be statistically significant, the practical magnitude of the differences is limited.

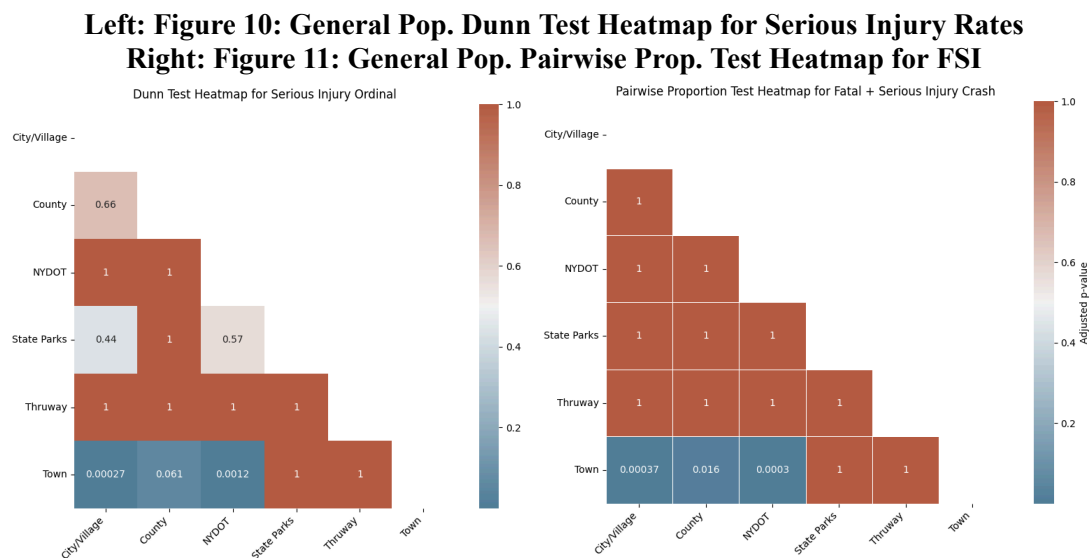
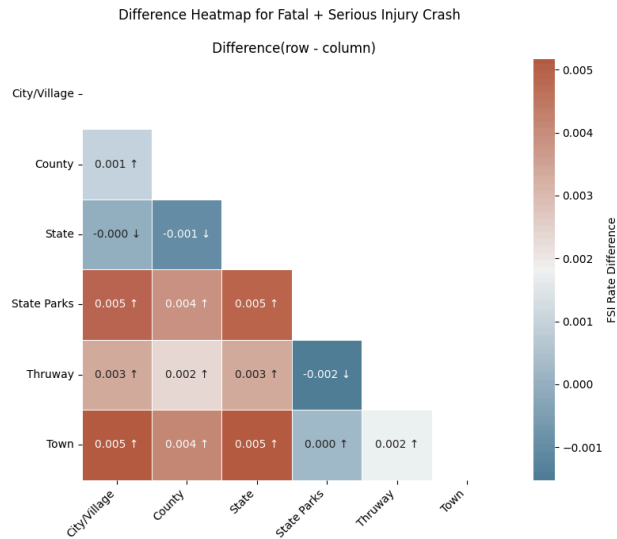


Figure 12: General Pop. Difference in FSI Rates for different jurisdictions



Results – Vulnerable Population

We then do a Chi-Squared test of independence for vulnerable populations between FSI and jurisdiction with:

H₀: FSI outcome is independent of jurisdiction in the Vulnerable population.

H_a: FSI outcome is not independent in the Vulnerable population.

The test result shows a statistic of ~ 22.54 and a p-value of 0.000412 with 5 degrees of freedom and small effect size with Cramér's $V = 0.06$. Therefore the results indicate a weak statistically significant association between jurisdiction and FSI outcome in the Vulnerable population in practical terms.

Thus, A pairwise-proportion z-test combined with Fisher's exact test on the FSI rate therefore were performed to compare FSI rates across jurisdictions in the Vulnerable population, with Bonferroni correction applied to adjust for multiple comparisons:

H₀: The two jurisdictions have the same FSI rate.

H_a: The two jurisdictions have different FSI rates.

The test shows statistically significant under a p-value of 0.05 where state (alias NYSDOT) -administered roads vs. county roads. According to the difference heatmap for FSI rate, there was a difference of approximately 5% for the FSI rate (figure 14). However, the large positive differences for State Parks should be interpreted with caution. Although the heatmap suggests a substantially higher FSI rate relative to several other jurisdictions, the State Parks group may have a relatively small sample size, which makes its estimated rate unstable and exaggerate pairwise differences.

For figure 15, it shows the result of a logistic regression comparing the log-odds of a crash being a fatal or serious one based on jurisdiction and whether you are a vulnerable or nonvulnerable user, with the reference group being a nonvulnerable road user driving on a city/village road. Being a vulnerable user

increases your odds of a crash being a fatal or serious one in a statistically significant manner, by ~11.23 times. In addition, controlling for vulnerability status, crashes on County roads have ~ 0.86 times the odds of being fatal or serious relative to crashes on City/Village roads, while crashes on Town roads have about 0.48 times the odds. Both of these effects are statistically significant. Within the vulnerable-user group, the predicted probability of an FSI crash is ~17.33% on City/Village roads, ~15.34% on County roads, and ~ 9.18% on Town roads. Relative to vulnerable crashes on City/Village roads, this corresponds to ~ 0.89 times the probability on County roads and ~ 0.53 times the probability on Town roads.

Left: Figure 13: Vulnerable Pop. Pairwise proportion for FSI over jurisdiction
Right: Figure 14: Vulnerable Pop. Difference in FSI Rates for different jurisdictions

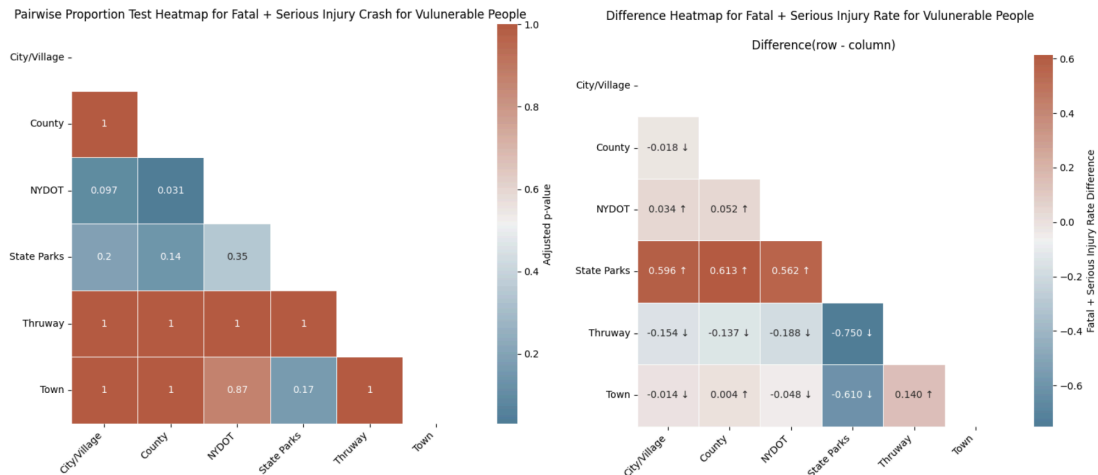


Figure 15: Logistic Regression; FSI Rate from vulnerability and Jurisdiction

Optimization terminated successfully.
Current function value: 0.099480
Iterations 9

Logit Regression Results						
Dep. Variable:	fsi_crash	No. Observations:	170280			
Model:	Logit	Df Residuals:	170273			
Method:	MLE	Df Model:	6			
Date:	Sat, 02 May 2026	Pseudo R-squ.:	0.05343			
Time:	04:37:23	Log-Likelihood:	-16940.			
converged:	True	LL-Null:	-17896.			
Covariance Type:	nonrobust	LLR p-value:	0.000			

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-3.9808	0.030	-132.091	0.000	-4.040	-3.922
C(VUN_group)[T.VUN]	2.4184	0.047	51.761	0.000	2.327	2.510
C(OWNING_JURISDICTION)[T.County]	-0.1460	0.052	-2.786	0.005	-0.249	-0.043
C(OWNING_JURISDICTION)[T.State]	0.0474	0.038	1.249	0.212	-0.027	0.122
C(OWNING_JURISDICTION)[T.State Parks]	-0.1630	0.176	-0.928	0.354	-0.507	0.181
C(OWNING_JURISDICTION)[T.Thruway]	-0.3552	0.382	-0.929	0.353	-1.105	0.394
C(OWNING_JURISDICTION)[T.Town]	-0.7292	0.200	-3.640	0.000	-1.122	-0.337

	odds_ratio	p_value
Intercept	0.018670	0.000000
C(VUN_group)[T.VUN]	11.227526	0.000000
C(OWNING_JURISDICTION)[T.County]	0.864150	0.005330
C(OWNING_JURISDICTION)[T.State]	1.048535	0.211802
C(OWNING_JURISDICTION)[T.State Parks]	0.849632	0.353583
C(OWNING_JURISDICTION)[T.Thruway]	0.701028	0.352929
C(OWNING_JURISDICTION)[T.Town]	0.482280	0.000272

5. Corridor Analysis

Previous work by Reconnect Rochester has uncovered ‘high-risk corridors’ or ‘crash corridors’ – specific sections of roadway that see elevated risks of fatalities and serious injuries. We were asked by Reconnect

Rochester to determine similar crash corridors from our data. In addition, we also examined the relationship between apparent crash factors & corridors.

Crash Corridor Creation

For our purposes, we defined corridors based on road infrastructure interactions: intersections, near intersections, and non-intersections. Intersection corridors are the parts of a road where two or more streets meet. Near-intersection corridors include the 100 feet of road leading up to an intersection. Non-intersection corridors are the remaining road segments between intersections, excluding those near-intersection areas.

The crashes were identified as belonging to a certain corridor based on their type of infrastructure interaction (IntersectionIndicator), street name (OnStreet), and closest cross street (ClosestCrossStreet). Along each road, crashes were projected onto a one-dimensional representation of the roadway to determine their relative position. This allowed us to group crashes into segments bounded by adjacent intersections. Once these corridors were defined, we estimated their lengths and general shapes using the spatial distribution of crashes within each segment.

Handling Incomplete Data

Our crash data was derived from police reports, and some records lacked the information needed to directly assign a corridor (missing or inconsistent street references). However, all crashes included geographic coordinates (UTM). To address this, we first used crashes with complete information to establish known corridors. These known crashes served as reference points for defining corridor locations and extents.

Non-intersection corridors were the only type with incomplete data. For these, we assigned corridor membership by identifying the nearest crash with known corridor information based on geographic proximity. The missing attributes were then inherited from this nearest neighbor.

Crash Corridor Ranking and Creation

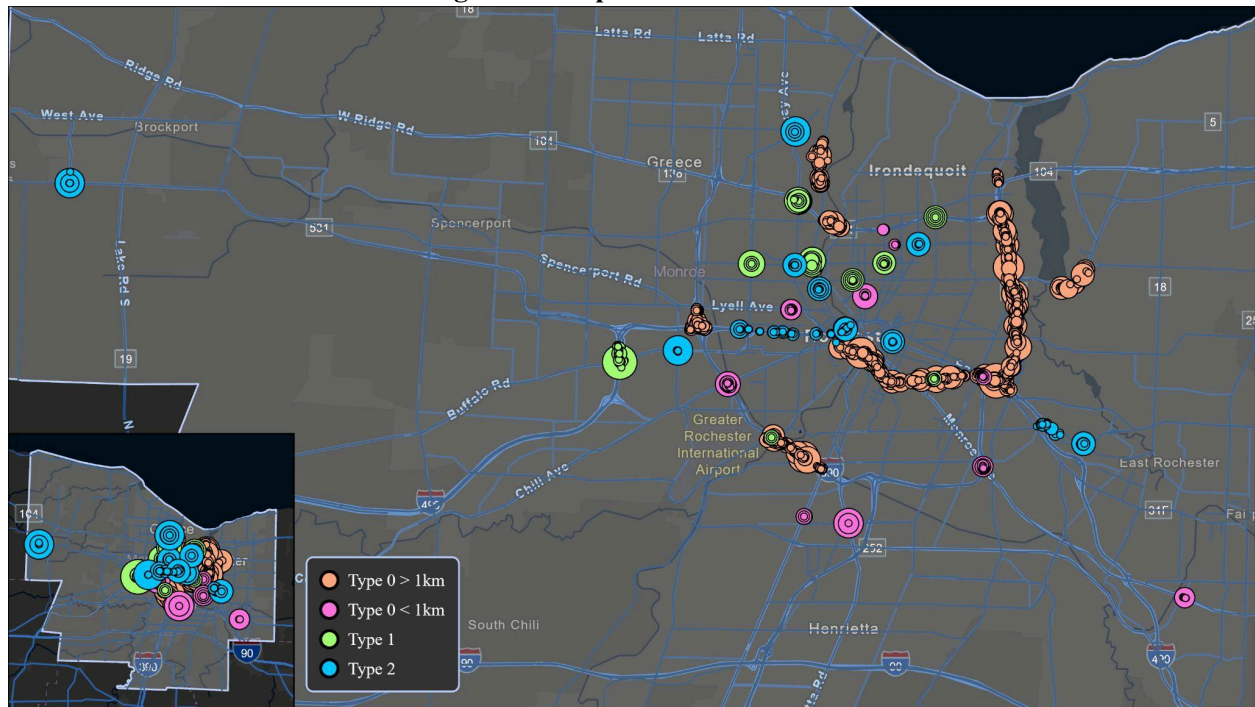
For each analysis category, corridors were first filtered based on intersection type and corridor length where applicable. This produced four distinct subsets: Intersection Type 0 (non-intersection) corridors split into those ≥ 1 km and < 1 km, as well as Intersection Types 1 (intersection) and 2 (near-intersection) considered separately. Within each subset, corridors were ranked based on the rate of serious injuries and fatalities per kilometer.

After establishing our sets of top corridors, we imported all four sets of top corridors to points with the XY table to point feature in ArcGIS, which permits importing a set of points to a map utilizing a coordinate system, using “UTMEasting” as the X coordinates and “UTMNorthing” for Y coordinates, thereby producing our maps.

We would like to note that crash corridor creation was the largest technical challenge for our project, taking up several weeks of time.

Figure 16 shows the resultant map with all top corridors.

Figure 16: Top Crash Corridors



Crash Corridor & Apparent Factors Data Extraction & EDA

Crash data was obtained from a standardized events dataset and filtered to include only observations within Monroe County using UTM coordinate bounds. Streets with fewer than 400 crashes were excluded to focus on high-frequency corridors. Crashes were categorized by type, with particular attention to vulnerable road users (pedestrians and bicyclists). The ApparentFactorCodes field was cleaned, split, and expanded to allow multiple contributing factors per crash to be analyzed individually, with invalid placeholder codes removed and valid codes mapped to interpretable categories.

Exploratory data analysis (EDA) examined the distribution of apparent factors across crash groups and roadway contexts. Proportional comparisons were visualized using stacked bar charts, and crashes were aggregated to the corridor level to compute metrics such as total fatalities and serious injuries (FSI) and FSI per kilometer. This provided both crash-level and corridor-level perspectives for identifying patterns in contributing factors and severity.

Crash Corridor & Apparent Factors Methodology

Crash events were grouped into roadway corridors, defined as segments between intersections. Intersection and near-intersection crashes were assigned fixed-length corridors, while non-intersection crashes were segmented using a data-driven approach based on Principal Component Analysis (PCA) to project crash locations along a roadway axis. Corridor lengths were then estimated geometrically, and crash severity was measured using Fatal and Serious Injuries (FSI), normalized by corridor length to allow comparison across segments.

Statistical methods were applied to evaluate relationships between apparent factors and crash types. Chi-square tests assessed whether factor distributions differed significantly across groups, while standardized residuals and odds ratios identified specific over- or under-represented factors. At the corridor level, piecewise linear regression models were used to examine how crash severity varies with

corridor length, with thresholds selected using AIC/BIC. Binomial generalized linear models (GLMs) were also used to model the likelihood of specific factors (e.g., Inattention, Failure to Yield) as a function of corridor characteristics.

Crash Corridor & Apparent Factors Results

The analysis identified clear differences in the distribution of apparent factors across crash groups. The most common factors in the general population were Inattention, Failure to Yield, and Unsafe Speed, while pedestrian and bicyclist crashes showed distinct patterns, with Failure to Yield being more strongly associated with bicyclist crashes and Inattention being more strongly associated with pedestrian crashes. A chi-square test produced a p-value of $p = 0.0004$, indicating that the relationship between apparent factors and crash type is statistically significant, with standardized residuals highlighting key overrepresented factors in each group.

At the corridor level, differences in crash severity were observed based on segment length. Corridors shorter than 1 km had higher median FSI compared to longer corridors, with a Mann-Whitney U test yielding $p = 1.896 \times 10^{-63}$. Piecewise regression identified a breakpoint at approximately 0.95 km, suggesting a change in how corridor length relates to crash severity. Factor-specific binomial models indicated that the likelihood of Inattention-related crashes does not significantly change with corridor length ($p = 0.708$), nor does Failure to Yield-related crashes ($p = 0.359$), providing no evidence that these behaviors are dependent on corridor length.

6. Road Project Analysis

Reconnect Rochester generally sponsors specific types of mitigations; in particular, it especially favors road diets, which rely on reducing the number of lanes in a road to reduce traffic & increase space for pedestrians and bicyclists [11]. As a result, they requested that we examine the effects of the East Avenue and Elmwood Avenue Road Diets; in addition, they also requested that we examine the Lake Avenue M&R and the Dewey & Driver Park Intersection Realignment.

Data Extraction

Road areas for the various projects, including rough construction periods, were given to us by Reconnect Rochester [12]. This was downloaded as a KML file and then loaded it to ArcGIS; we then used ‘select by location’ to identify and associate corridor points within 2 miles of a road project, and using this information to create a new dataset (“RoadProjectCrashes.csv”). The associated road projects were saved to the column “RoadProject”.

Methodology

We used a metric based on the methodology used previously by the Minnesota Department of Transportation [13] to analyze roads: namely, we used a paired Wilcoxon signed-rank test [14] to compare monthly crashes, as well as fatal and serious injuries, in the year prior to and immediately after the project was constructed. For example, if a road was rebuilt 2022-2023, we would compare monthly fatal and serious injury counts (for all users) and crashes in 2021 and 2024. This is achieved by taking the sum of all fatal and serious injuries for each month, in each year, and then taking the difference, before carrying out a Wilcoxon signed-rank test on that difference – testing both for a statistically significant decrease and a statistically significant increase, with a significance level of 0.05.

The Wilcoxon, as a non-parametric test, permits us to avoid issues with the non-normal distribution of fatalities and serious injuries [14].

East Avenue Road Diet

A section of East Avenue was placed on a road diet around 2018-2019, reducing the four-lane road to two travel lanes and a turn lane, as well as adding shoulders to protect bicyclists and pedestrians [15] [16].

We compared the years 2017 and 2020 through the Wilcoxon signed-rank test. Testing for a decrease in FSI and total crashes returned p-values of 0.5 for FSI & ~0.0015 for total crashes. We therefore find a statistically significant decrease in total crashes – but not for fatal and serious injuries. Subsequently, we tested for an increase in FSI, which returned a p-value of 1.0.

This implies that there is no statistically significant change in all in FSI, but a statistically significant decrease in total crashes. *However*, this does not account for COVID-19.

Elmwood Avenue Road Diet

The South Avenue and Elmwood Avenue Improvement Project (or the Elmwood Avenue Road Diet) was constructed in Spring 2022 through Fall 2023. It narrowed Elmwood Avenue & added bike lanes, sidewalks, and pedestrian crossing improvements.[17]

We compared the years 2021 and 2024 through the Wilcoxon signed-rank test.

Testing for a decrease in FSI and total crashes returned p-values of 0.75 for FSI & ~0.944 for total crashes. In both cases, we fail to reject the null hypothesis, and find no statistically significant decrease in either total crashes or fatal and serious injury counts.

Testing for an increase in FSI and total crashes returned p-values of 0.5 for FSI & ~0.0745 for total crashes. In the case of FSI, we fail to reject the null hypothesis, and find no statistically significant increase in fatal and serious injury counts. For total crashes, the p-value of ~0.0745 is relatively low, but still above a confidence level of ~0.05, so we also fail to reject the null hypothesis.

Lake Avenue M&R

The Lake Avenue Milling and Resurfacing was a construction project which resurfaced Lake Avenue along a 1.3 mile stretch bordering Holy Sepulchre Cemetery. Unlike geometric redesign projects such as road diets or intersection realignments, this intervention primarily focused on pavement resurfacing and structural road quality improvements, with no major changes to lane configuration or traffic control design.

We compared the years 2015 and 2017 using the Wilcoxon signed-rank test, following the same monthly aggregation approach used in previous project evaluations. As with prior analyses, we tested both for a statistically significant decrease and increase in fatal and serious injuries (FSI) and total crashes. Testing for a decrease in FSI and total crashes returned p-values of approximately FSI $p = .75$ and crashes $p = .752$ respectively. In both cases, we fail to reject the null hypothesis, indicating no statistically significant reduction in either fatal and serious injuries or total crash counts following the intervention.

Testing for an increase in FSI and total crashes returned p-values of approximately FSI $p = 0.75$ and crashes $p = \sim 0.287$ respectively. Again, in both cases we fail to reject the null hypothesis, suggesting no statistically significant increase in safety-related outcomes. Overall, the Lake Avenue M&R project does not appear to produce a measurable change in crash frequency or severity within the limits of this aggregated before-and-after comparison framework.

Dewey & Driver Park Intersection Realignment

The Dewey & Driver Park Intersection Realignment was carried out from around Spring 2017 to Summer 2018. It realigned the offset intersection at Dewey & Driver Park, with the intent of improving traffic flow and reducing crashes; in addition, it provided for new sidewalks [18].

We compared the years 2016 and 2019 through the Wilcoxon signed-rank test.

Testing for a decrease in FSI and total crashes returned p-values of 0.75 for FSI & ~ 0.254 for total crashes. In both cases, we fail to reject the null hypothesis, and find no statistically significant decrease in either total crashes or fatal and serious injury counts.

Testing for an increase in FSI and total crashes returned p-values of 0.75 for FSI & ~ 0.761 for total crashes. In both cases, we fail to reject the null hypothesis, and find no statistically significant increase in either total crashes or fatal and serious injury counts.

7. Conclusions & Future Steps

We have found *initial and very preliminary* results to the four primary questions posed to us by Reconnect Rochester.

We have identified that vulnerable road users are more likely to die if hit later at night, from around 20:00 to 06:00. We have established that there are small but statistically significant differences in crash outcomes across jurisdictions, especially with regards to crashes that happen on state / NYDOT roads. We have identified a number of corridors for heightened FSI, both for the general population & vulnerable user population.

We have examined, briefly, a number of road projects, and an initial, naive analysis has not found statistically significant outcomes, save for the East Avenue road diet (which, to reiterate, might be confounded with COVID-19).

However, a number of factors were not accounted for in our analysis. We did not account for any potential change in definitions for Serious Injuries / Injuries. More pressingly, as stated before, we did not account for the result of changes resulting from the COVID-19 pandemic, which is especially crucial to note for our road project analysis. Any future analysis should take these factors into account if at all possible.

In addition, any future analysis, especially for the effects of road projects, could benefit greatly from the addition of traffic flow data, which we did not have access to for this analysis. Adding traffic flow data would permit the use of more powerful techniques such as an Expected Bayes model [15] to analyze the effectiveness of road improvements, and could potentially help to normalize data in such a manner as to reduce the effect of COVID-19. It is also worth noting that for projects such as the Elmwood Avenue Road Diet, more time may be required to determine the results, due to the rarity of crashes & fatal / serious injuries.

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